UNIT 4 — MORPHOLOGY OF FLOWERING PLANTS

Study notes based on the supplied textbook pages, expanded with useful identification points and examples for competitive examination practice.

1. PLANT BODY ORGANISATION

• **Plant morphology** is the study of the external form and visible parts of plants.

• A typical plant body has two main systems: the **root system** (usually below the ground) and the **shoot system** (usually above the ground).

• The shoot system includes the stem, branches, leaves, buds, flowers and fruits.

Plant form	General characteristics	Examples
Herbs	Small plants with soft, green stems.	Mint, coriander, wheat.
Shrubs	Medium-sized, bushy plants with hard stems and branches near the base.	Rose, hibiscus, cotton.
Trees	Tall plants with a thick, woody main stem or trunk.	Mango, neem, banyan.
Climbers	Weak-stemmed plants that need support and climb upwards.	Grapevine, pea, money plant.
Creepers	Weak-stemmed plants that spread along the ground.	Pumpkin, watermelon, cucumber.

2. ROOT SYSTEM

• The root is generally the underground, non-green part of a plant. It anchors the plant and absorbs water and minerals from the soil.

• Roots bind soil particles and help reduce soil erosion. In some plants, roots store reserve food.

Main root systems

Taproot system	Fibrous root system
One main root develops from the radicle and grows downwards. Smaller lateral roots arise from it.	Many thin roots of similar size arise in a cluster from the base of the stem.
Common in many dicots.	Common in many monocots.
Examples: mustard, mango, neem, carrot.	Examples: wheat, maize, rice, grasses.

Root modifications

• **Storage roots:** roots become swollen and store food; examples include carrot, turnip and sweet potato.

• **Supporting roots:** prop roots of banyan grow from branches and support them; stilt roots arise from lower stem nodes, as in sugarcane.

• **Clinging/climbing roots:** help plants attach to walls or supports; examples include betel and black pepper.

• Do not confuse a potato tuber with a root: potato is a modified underground **stem**, whereas carrot and sweet potato are storage roots.

3. STEM

• The stem usually develops from the plumule. It bears branches, leaves, buds, flowers and fruits and connects the roots with the rest of the plant.

• Functions: supports the aerial parts, holds leaves in positions where they receive light, and transports water, minerals and prepared food through conducting tissues.

Stem modifications

Modification	Function and examples
Food storage	Underground stems store food: potato (tuber), ginger and turmeric (rhizomes).
Photosynthesis	Green, succulent stems make food in plants with reduced leaves, e.g. cactus.
Climbing	Stem tendrils coil around supports, e.g. grapevine.
Vegetative propagation	New plants can grow from stem cuttings, e.g. rose, jasmine and hibiscus.
Protection	Thorns or prickles help protect plants, e.g. bougainvillea and rose.
Water storage	Fleshy stems store water, e.g. cactus.

4. LEAF

• A leaf is usually a flattened, green structure arising from a stem or branch. It is the main organ of photosynthesis in most plants.

• The broad, flat part is the **lamina (leaf blade)**. The **petiole** is the stalk attaching the leaf to the stem. The leaf base attaches the leaf to the stem; stipules may be present at the base in some plants.

• Leaf apex = tip of the leaf; leaf margin = edge; midrib = main central vein; veins branch from the midrib.

Venation

Reticulate venation	Parallel venation
Veins form a net-like pattern.	Veins run more or less parallel to one another.
Commonly associated with dicot plants.	Commonly associated with monocot plants.
Examples: mango, hibiscus, peepal.	Examples: banana, wheat, maize.

Functions and modifications of leaves

• **Photosynthesis:** leaves use carbon dioxide and water in the presence of sunlight and chlorophyll to manufacture food; oxygen is released.

• **Transpiration:** loss of water vapour, mainly through stomata on leaf surfaces.

• **Support:** leaf tendrils help climbing, e.g. pea.

• **Food storage:** fleshy leaves store food; edible leaves include spinach, cabbage and lettuce.

• **Vegetative reproduction:** buds on leaf margins can develop into plantlets, e.g. Bryophyllum.

• **Protection:** leaves may become spines, as in cactus, reducing water loss and protecting the plant.

5. FLOWER

• The flower is the reproductive structure of flowering plants. It develops from a flower bud and helps in sexual reproduction.

• A typical flower has four whorls arranged from the outside towards the centre: **calyx, corolla, androecium and gynoecium**.

Whorl	Main parts	Function
Calyx	Sepals	Usually green; protect the flower in the bud stage.
Corolla	Petals	Often coloured or scented; attract pollinators.
Androecium	Stamens (male part)	Produces pollen grains. Each typical stamen has anther and filament.
Gynoecium	Carpel/pistil (female part)	Includes stigma, style and ovary; ovary contains ovules.

Important flower-part details

• **Anther:** part of the stamen that produces pollen grains. **Filament:** stalk supporting the anther.

• **Stigma:** receives pollen. **Style:** connects stigma to ovary. **Ovary:** contains ovules; after fertilisation, ovules develop into seeds and the ovary usually develops into a fruit.

• Flowers may provide nectar to insects; pollinators help transfer pollen between flowers.

6. HIGH-VALUE COMPARISONS

Feature	Typical monocot pattern	Typical dicot pattern
Cotyledons	One	Two
Root system	Usually fibrous	Usually taproot
Leaf venation	Usually parallel	Usually reticulate
Floral parts	Often in multiples of 3	Often in multiples of 4 or 5

These are common patterns, not absolute rules for every plant. Use the actual features given in a question rather than assuming one feature proves every other feature.

7. EXAM QUICK RECALL

• Carrot and turnip: storage roots. Potato: modified stem tuber. Ginger and turmeric: rhizomes.

• Banyan: prop roots. Sugarcane: stilt roots. Pea: leaf tendrils. Grapevine: stem tendrils. Bryophyllum: leaf buds.

• Reticulate = net-like veins; parallel = side-by-side veins.

• Sepals protect; petals attract; stamens are male; carpel/pistil is female.

• Stamen = anther + filament. Pistil/carpel = stigma + style + ovary.

• Flower is the reproductive organ; after fertilisation, ovules usually become seeds and ovary usually becomes fruit.

Revise the examples carefully. Homi Bhabha questions often test the function of a plant part, the modification responsible for it, and the correct example.